

How to Win Consumer Intention in E-Commerce Platform in a High-Cost Urban City: A Case Study of Hanoi

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ABSTRACT

This study investigated factors that influence online shopping intentions in Hanoi, Vietnam, a rapidly growing urban center with a high cost of living. Using an enhanced Unified Theory of Acceptance and Use of Technology 2 model and survey data from 266 respondents, it examined the impact of electronic word of mouth, hedonic motivation, price value, and perceived risk. Electronic word of mouth emerged as the most significant driver, followed by hedonic value and the perception of e-commerce as an efficient alternative to traditional retail. Lower income consumers showed higher online shopping intentions than higher income groups, challenging conventional assumptions. This study contributes to the literature by highlighting how these factors uniquely operate in high-cost urban contexts. It offers actionable insights for e-commerce platforms, emphasizing the importance of cultivating positive reviews, enhancing shopping experiences, and developing strategies tailored to economically constrained consumers in competitive urban markets.

KEYWORDS

Consumer Behavior, Electronic Word-of-Mouth, Long-Tail Effect, Online Purchase Intention, UTAUT2 Model

INTRODUCTION

Rapid advancements in digital technology have positioned *e-commerce*—defined as the purchase of products or services via the internet—as a central component of contemporary consumer behavior. E-commerce offers significant advantages over traditional retail, including convenience, accessibility, flexible shopping times, and a wider range of product choices (Jiang et al., 2013; Ozen & Engizek,

DOI: 10.4018/IJABIM.377607

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2014; Wu et al., 2011). Such benefits are especially valuable in urban areas, where busy lifestyles and limited time amplify the appeal of online shopping (Monoarfa et al., 2024). Online platforms also enhance consumer engagement by providing targeted promotions, competitive pricing, and personalized product recommendations, which collectively build consumer trust and facilitate informed decision making (Armstrong, 2008; Harris-Lagoudakis, 2023).

Despite these benefits, however, online shopping presents distinct challenges, in particular in high-cost urban settings. Delivery fees, potential delays, and uncertainties related to product quality and customer service disproportionately affect lower income consumers, causing increased caution toward the full embracement of online purchasing (Hebbar et al., 2020). Therefore, understanding *online shopping intention*—defined as consumers' willingness and readiness to engage in digital transactions (Pavlou, 2003; Zwass, 1999)—becomes crucial. Online shopping intention directly reflects consumers' desires, anticipations, and readiness to make purchases online (Ajzen, 1991; Armitage & Conner, 2001; Blackwell et al., 2002; Ghosh, 1990; Magesh, 2011; Mirabi et al., 2015; Prestwich et al., 2008; Raza et al., 2014; Salisbury et al., 2001; Shah et al., 2012; Tirtiroglu & Elbeck, 2008). Investigating the determinants of this intention within economically constrained urban contexts is therefore essential for understanding how consumers navigate digital marketplaces, manage perceived risks, and capitalize on the comparative advantages provided by e-commerce platforms (Hasslinger et al., 2007; Sindhav & Balazs, 1999; Vrechopoulos et al., 2001; Ward & Lee, 2000).

E-commerce plays a pivotal role in Vietnam's digital economy, placing the country among Southeast Asia's fastest growing online markets (Nam, 2023; Nhi, 2024). Online shoppers now account for 74.8% of the population, with an average annual spend of USD 260–285 (Ministry of Trade, 2022). Although early platforms, such as Lazada and Tiki, have seen market share declines, Shopee and TikTok Shop have rapidly expanded by using strategies such as shoppertainment (shopping combined with entertainment activities like livestreaming and interactive games) and aggressive promotions (Ha Linh, 2024). Nevertheless, Vietnam's e-commerce sector, like to others in Southeast Asia, faces growing challenges, including changing consumer and merchant behaviors, a broader product portfolio, and logistical complexities (Arora et al., 2025). Consumers display an increasingly cautious but strong preference for affordable, quality products, reflecting a regional focus on value (PWC, 2023; Yendamuri et al., 2023; PWC, 2023). At the same time, government initiatives encourage deeper digital engagement among small- and medium-size enterprises and mobile channels, underscoring the sector's importance to economic progress (Ngoc, 2022).

Within this national setting, Hanoi exemplifies both rapid economic development and mounting living costs. For 5 consecutive years, it has been recognized as Vietnam's most expensive city, marked by elevated expenses in housing, transportation, and essential services (van der Heijden, 2004). These financial strains weigh heavily on residents—in particular, those earning <USD 80/month—who struggle to meet basic needs (Duranton & Puga, 2014; Fernández & Rubiera, 2017; Navamuel et al., 2019; Timmins, 2006; Wu et al., 2011). In response, the local government introduced the 2024 e-commerce development plan, aiming to increase online participation and expand business-to-consumer e-commerce to 13% of the total number of retail enterprises (Ngoc, 2022).

This study aimed to investigate the main motivations and barriers that influence consumers' online shopping intentions in Hanoi. To be specific, it explored whether e-commerce offers a viable alternative to traditional shopping, enabling consumers—in particular, those with lower incomes—to access affordable goods and services despite rising living costs. Understanding these dynamics is vital for developing effective strategies that address the needs of economically constrained consumers in urban environments. This study also examined how businesses can capture consumer interest and shape online shopping intentions in this high-cost urban context, addressing the priorities of policymakers and industry stakeholders.

This study applied an enhanced version of the Unified Theory of Acceptance and Use of Technology 2 (UTAUT2) model, incorporating factors such as electronic word of mouth (eWOM) and the long-tail effect, to assess their impact on consumer behavior in the context of high-cost

urban areas. By addressing gaps in understanding how economic constraints affect online shopping in cities with high living costs, this study provides valuable insights that can be applied to similar urban settings across Southeast Asia.

This article is organized to thoroughly address the research objectives. After the introduction, the theoretical framework and hypotheses are presented, and the relevant theories and the hypotheses to be tested are discussed. In the “Research Methodology” section, the research design, data collection methods, and analysis techniques are outlined. The findings are presented in the “Results” section. This is followed by a “Discussion” section that connects the results to the theoretical framework and existing literature. In the “Conclusion” section the key findings and their implications for e-commerce strategies are summarized, and recommendations for future research are offered.

THEORETICAL FRAMEWORK AND HYPOTHESIS DEVELOPMENT

Theoretical Framework

Several frameworks have examined determinants of online purchase intentions. Early models include the theory of reasoned action (Ajzen & Fishbein, 1975) and the theory of planned behavior (Ajzen, 1991), both of which highlight attitudes, subjective norms, and perceived behavioral control. The technology acceptance model, introduced by Davis (1986), refined the significance of perceived usefulness and ease of use, later evolving into combined models such as Combined Technology Acceptance Model and Theory of Planned Behavior (C-TAM-TPB) (Taylor & Todd, 1995) and Extended Consumer Acceptance Model (e-CAM) (Lohse et al., 2000). The UTAUT (Venkatesh et al., 2003) synthesized these approaches and introduced constructs such as effort expectancy and social influence, and UTAUT2 (Venkatesh et al., 2012) added hedonic motivation and price value. Although UTAUT is widely applied, including in Vietnam (Singh, 2020), empirical research on UTAUT2 remains limited.

Nevertheless, UTAUT2 alone may not fully capture how heightened living costs and economic constraints amplify perceived risk in high-cost urban environments. The theory of perceived risk, developed by Bauer (1960), underscores how uncertainty about product or service quality and transaction security can deter online shoppers—factors that grow more critical where financial vulnerability intensifies consumer caution. By integrating the theory of perceived risk with UTAUT2’s core dimensions (e.g., efficiency expectation, effort expectancy, social influence, and hedonic motivation), and including price value, the long-tail effect, and eWOM, this study proposes a unified framework to address both motivational and risk-based considerations. Hence, in a response to calls for deeper exploration of how economic burdens shape consumers’ digital behavior, efficiency expectation, social influence, price value, hedonic motivation, the long-tail effect, eWOM, and perceived risk are posited as seven main factors that influence online shopping intentions in high-cost urban settings. These factors, supported by existing research, are presented in Table 1, which outlines the relevant constructs and highlights previous studies that have examined their impact on online shopping intentions.

Table 1. Key factors that influence online shopping intentions in high-cost urban cities

Factor	Studies
Efficiency expectation	Venkatesh et al. (2012), Hallikainen et al. (2022), Carson et al. (1996), Angelina et al. (2022), Singh & Matsui (2017), Dong et al. (2020), Zhao & Wang (2015), Nguyen et al. (2019), Doan (2020)
Social influence	Venkatesh et al. (2012), Gerpott (2010), Alshehri & Meziane (2015), Angelina et al. (2022), Singh & Matsui (2017), Zhang et al. (2023), Nguyen et al. (2019), Doan (2020)

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Table 1. Continued

Factor	Studies
Price value	Dodds et al. (1991), Venkatesh et al. (2012), Nguyen et al. (2019), Singh & Matsui (2017)
Hedonic motivation	van der Heijden (2004), Brown & Venkatesh (2005), Brynjolfsson et al. (2011), Park et al. (2011), Wu et al. (2011), M. Singh & Matsui (2017), Monoarfa et al. (2024b), Phelps et al. (2004)
The long-tail effect	Nguyen et al. (2019), Singh & Matsui (2017)
Electronic word of mouth	Phelps et al., (2004), Park et al., (2011), Mayer et al. (1995), Javadi et al. (2012), Jarvenpaa et al. (2000), Monsuwé et al. (2004), Dinev & Hart (2006)
Perceived risk	Bauer (1960), Chiu et al. (2012), Mayer et al. (1995), Javadi et al. (2012), Jarvenpaa et al. (2000), Monsuwé et al. (2004), Dinev & Hart (2006), Wagner et al. (2019), Hebbar et al. (2020), Cheng et al. (2002), Doan (2020), Nguyen (2023)

Hypothesis Development

Building on the theoretical framework seven key constructs—efficiency expectation, social influence, price value, hedonic motivation, the long-tail effect, eWOM, and perceived risk—are examined to determine how consumers form online shopping intentions in high-cost urban environments.

Efficiency expectations in the context of e-commerce shopping refers to the extent to which consumers believe that advanced technological systems on online platforms can facilitate their consumption experiences (Venkatesh et al., 2012). These expectations include benefits such as saving time (Carson et al., 1996; Hebbar et al., 2020; Nguyen et al., 2019), reducing fatigue (Carson et al., 1996), enabling shopping at any time (Carson et al., 1996; Hebbar et al., 2020), facilitating the search for product information (Carson et al., 1996), saving shopping costs, allowing price comparisons (Carson et al., 1996), and providing convenient delivery services. In addition, consumers can further reduce expenses by using online discount codes to save on shipping costs and other related charges. The combination of high population density, limited physical shopping options, and significant opportunity costs makes efficiency expectations a key driver of online shopping behavior in high-cost urban settings (Cheung & Thadani, 2012).

Hypothesis 1: Efficiency expectation positively influences consumers' intentions to shop online.

Social influence refers to the extent to which customers perceive the importance of other consumers' beliefs in adopting new technology (Monoarfa et al., 2024) or, more specifically, the impact of family, friends, colleagues, and other social connections on their decisions regarding online shopping (Gerpott, 2010). High-cost urban environments often feature diverse and interconnected social networks, which amplify the reach and intensity of social influence on consumer behavior (Sepac et al., 2024). Consumers in high-cost urban areas rely heavily on these social sources to guide their purchasing decisions, in particular for products involving significant financial commitments (Nguyen et al., 2019). They are also highly responsive to social validation and endorsements from influencers and trusted sources (Lee & Youn, 2009). In addition, these consumers may feel pressure to adopt trendy products or follow recommendations to enhance social acceptance (Cheung & Thadani, 2012; Zhang et al., 2023).

Hypothesis 2: Social influence positively influences consumers' intention to shop online.

Price value is the consideration between the benefits received from online shopping and the costs incurred to carry out that shopping. If the benefits of online shopping outweigh the costs, then price

value will have a positive impact. The availability of discounts; promotions; competitive pricing; and additional benefits (e.g., free shipping, loyalty rewards, and flexible payment options on e-commerce platforms) enhances the perceived price value for consumers. This is particularly true for high-cost urban consumers because they aim to save as much as possible, optimize their expenditures (Chen et al., 2004; Nguyen et al., 2019), and seek more value-added benefits (Lee & Youn, 2009). However, this also means that consumers cannot inspect products directly before purchasing, and the price value then depends on consumers' trust in the product and the business's brand. According to Jiang et al. (2013), price is often a necessary criterion for consumers to assess product quality when shopping online because customers cannot physically inspect products before purchasing.

Hypothesis 3: Price value positively influences consumers' intention of online shopping.

Hedonic motivation is the pleasure or interest derived from online shopping activities; it has been shown to play an essential role in the acceptance and use of technology (Brown & Venkatesh, 2005). In information systems, hedonic motivation directly influences the acceptance and use of technology (Gerpott, 2010). In the context of consumers, hedonic motivation has also been demonstrated to be an essential factor in the decision to accept and use technology (Brown & Venkatesh, 2005; Carson et al., 1996). For some consumers, using a new and innovative shopping method can bring excitement and entertainment. In high-cost urban environments like Hanoi, consumers often turn to hedonic shopping experiences as a way to relieve stress and gain emotional satisfaction. In addition, urban consumers with higher disposable incomes and greater exposure to digital trends are more likely to prioritize hedonic motivations in their online shopping behavior (Zhang et al., 2023).

Hypothesis 4: Hedonic motivation positively influences consumers' intention to shop online.

The *long-tail effect* describes how businesses leverage e-commerce platforms to generate significant revenue from a wide array of niche products. Digital platforms cater to diverse consumer preferences (Anderson, 2014) by leveraging advanced recommendation systems and search algorithms to expose consumers to less mainstream options (Brynjolfsson et al., 2011) and personalize goods recommendations that significantly enhance their willingness to shop online (Singh & Matsui, 2017). In high-cost urban environments like Hanoi, consumers value both varieties and convenience, which help them address time constraints; therefore, they tend to prefer platforms that offer extensive product ranges (Jiang et al., 2013). In addition, the long-tail effect helps address financial pressures by providing affordable niche options.

Hypothesis 5: The long-tail effect positively influences consumers' online shopping intention.

EWOM refers to consumer-generated content, such as online reviews, testimonials, and ratings, that significantly influences online shopping behavior. It reduces uncertainty and helps customers make informed purchase decisions, in particular on e-commerce platforms (Cheung & Thadani, 2012; Lee & Youn, 2009; Park et al., 2011; Phelps et al., 2004; Wu et al., 2011; Zhu & Kraemer, 2005). In high-cost urban areas, consumers value detailed feedback that addresses their diverse needs, making high-quality eWOM—characterized by clarity, reliability, and verified reviews—especially influential in shaping online shopping intentions (Cheung & Thadani, 2012; Nguyen et al., 2019). Such feedback helps reduce perceived financial risks, which are a major concern for these consumers. Positive eWOM enhances purchase intentions, whereas negative eWOM can significantly deter them (Sepac et al., 2024). However, in high-cost urban environments consumers critically evaluate negative feedback because of the substantial financial stakes involved in their purchasing decisions.

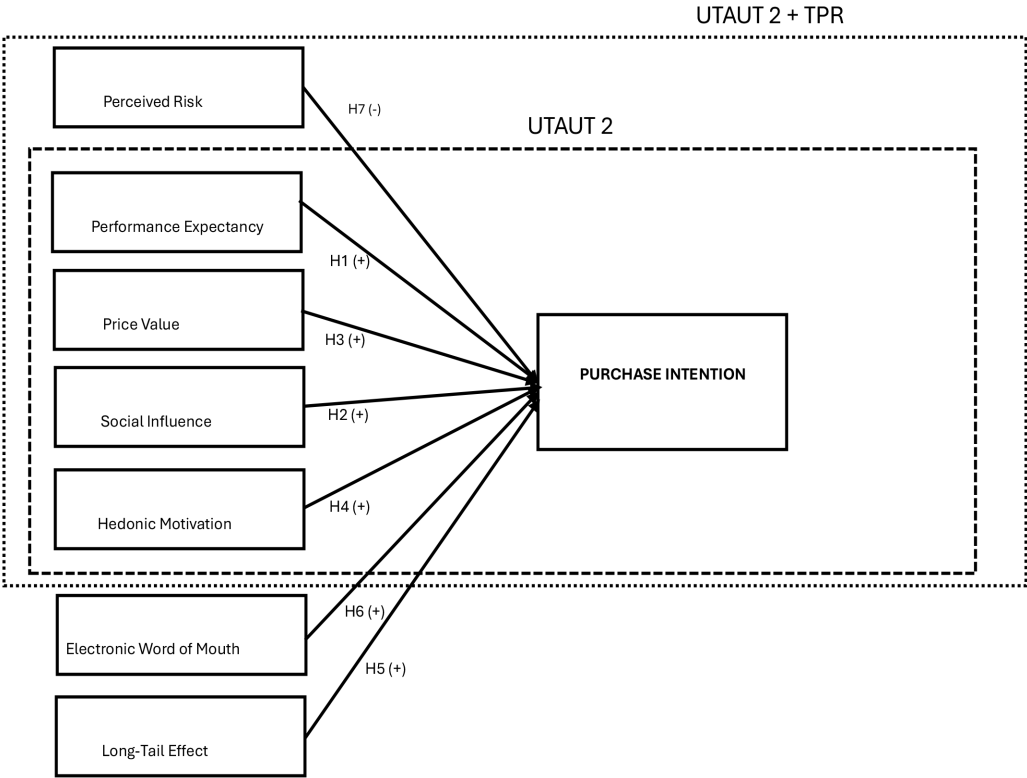
Hypothesis 6: EWOM positively influences consumers' online shopping intention.

Perceived risk refers to the uncertainty and potential negative consequences consumers associate with online shopping (Mayer et al., 1995). This concept is multidimensional, encompassing financial (Chen et al., 2004), product (Nguyen et al., 2019), security (Zhang et al., 2023), time (Lee & Youn, 2009), social, and psychological risks (Sepac et al., 2024). Each dimension uniquely influences customer behavior, in particular in e-commerce settings (Ariffin et al., 2018). Among these dimensions, financial risk is often amplified in high-cost urban areas, where the stakes of financial loss are significantly higher (Chen et al., 2004). In addition, the inability to physically inspect products online contributes to product risk, creating uncertainties about quality and performance. This is especially problematic for high-cost urban consumers, who typically have higher expectations of quality (Nguyen et al., 2019). Other risk dimensions are similarly intensified in high-cost urban environments. Security risks, such as concerns over payment security and data privacy, are particularly heightened (Zhang et al., 2023). Time risks, including delays in delivery and challenges with returns, further deter online shopping (Lee & Youn, 2009). Social acceptance concerns and psychological discomfort (Sepac et al., 2024) also play a role in shaping consumer behavior. A lack of trust is one of the primary barriers that prevents consumers from engaging in e-commerce (Jarvenpaa et al., 2000; Monsuwé et al., 2004). Perceived uncertainty in the online environment often causes hesitation, making individuals less likely to conduct online transactions (Dinev & Hart, 2006).

Hypothesis 7: Perceived risk negatively influences consumers' online shopping intentions.

The framework depicted in Figure 1 evaluates and analyzes the factors influencing consumers' intention to shop online in Hanoi within the unique context of e-commerce platforms in a high-cost urban landscape.

Figure 1. Proposed research model and hypothesis



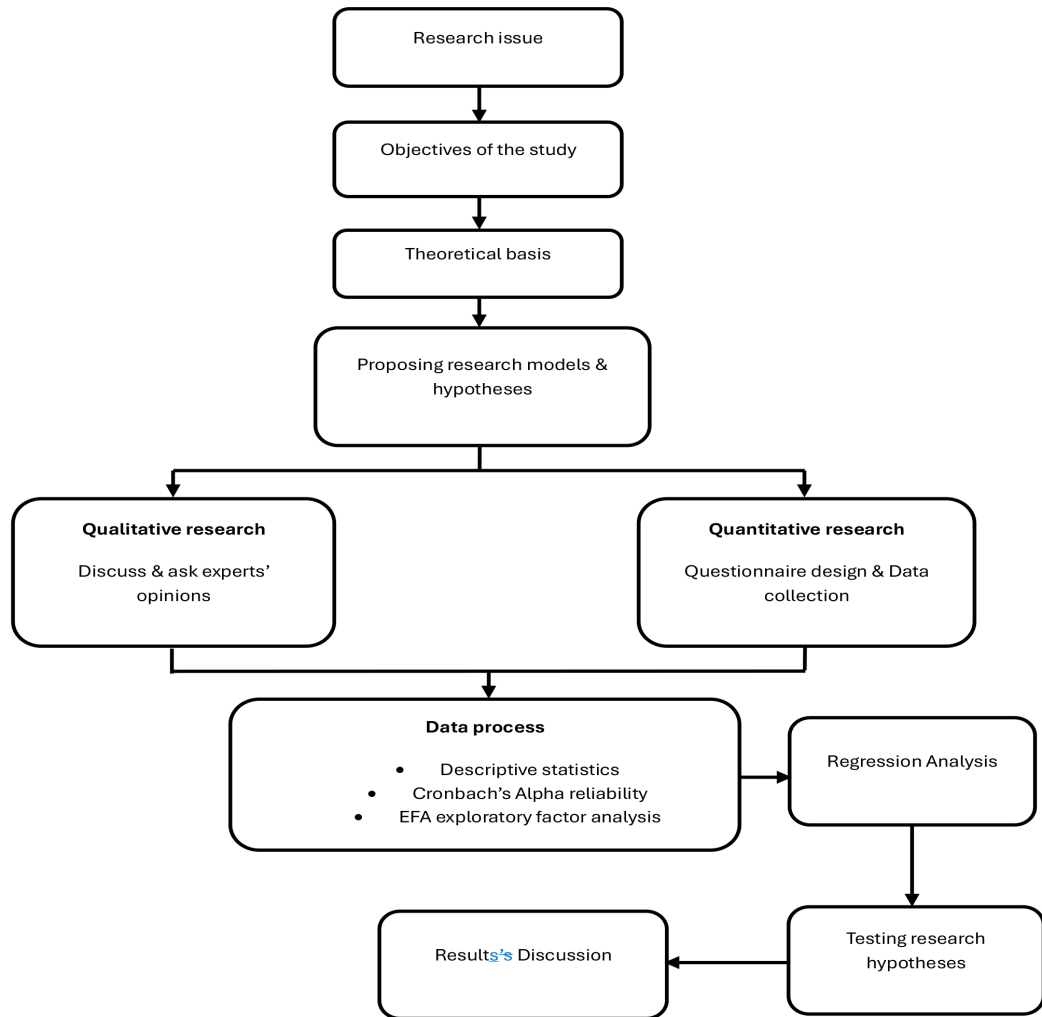
Note. UTAUT 2 = Unified Theory of Acceptance and Use of Technology; TPR = theory of perceived risk; H = hypothesis.

RESEARCH METHODOLOGY

Research Procedure and Data Collection

This research used a mixed-methods approach consisting of two phases (Figure 2): a preliminary qualitative phase followed by a formal quantitative phase.

Figure 2. Research procedure



In the qualitative phase of this study, expert consultations and focus groups were conducted with specialists in economics and experienced online consumers from academic and corporate sectors. These sessions aimed to refine the research model and verify the practical relevance and theoretical appropriateness of the identified influencing factors. The study was conducted in accordance with widely accepted ethical research standards, including the principles outlined in the Declaration of Helsinki and the International Chamber of Commerce/European Society for Opinion and Marketing Research (ICC/ESOMAR) guidelines for social and market research. Informed consent was obtained from all participants, and procedures followed best practices for voluntary participation, data confidentiality, and the responsible handling of personal information.

The quantitative phase of this study involved developing and administering a structured questionnaire that was based on validated scales from the existing literature and refined insights from the qualitative discussions. The questionnaire was designed to ensure both theoretical alignment and contextual relevance. A pilot test was conducted with a small group of participants to assess the clarity, reliability, and logical flow of the items. On the basis of the pilot test feedback, the

questionnaire was revised to improve readability and measurement accuracy. The final instrument comprised three sections: (a) screening questions to confirm prior online shopping experience; (b) a set of items that measured seven key constructs—efficiency expectation, social influence, price value, hedonic motivation, long-tail effect, eWOM, and perceived risk—using a 5-point Likert scale; and (c) demographic questions that inquired about age, gender, occupation, and monthly income. Convenience sampling, a nonprobability sampling method, was chosen because of its practicality, cost-effectiveness, and accessibility. The minimum sample size required was determined by following established guidelines: a minimum of 120 respondents for exploratory factor analysis (at least five times the number of observed variables) and 106 respondents for regression analysis ($N \geq 50 + 8 \times p$, where p is the number of independent variables). Two hundred seventy-six questionnaires were distributed to accommodate potential invalid responses. After data validation and removal of incomplete or logically inconsistent questionnaires, the final dataset comprised 266 valid responses.

Respondent demographics were diverse, encompassing nearly equal gender representation and ages predominantly between 18 and 40 years. With respect to occupation, respondents included office workers (32%), general laborers (31.2%), freelancers (22.6%), and students (14.3%). The majority (73.3%) reported monthly incomes ranging from USD 80 to 200, representing a typical cross-section of consumers in high-cost urban areas.

Data Analysis

Quantitative analysis were done using SPSS (Version 23) software, following a systematic and rigorous analytical process:

- *Data preparation and descriptive statistics.* Data screening and descriptive analysis (mean, frequency, and standard deviation) ensured clarity of respondent profiles.
- *Reliability analysis (Cronbach's α):* The internal consistency of measurement scales was evaluated with a Cronbach's α threshold of $\geq .6$.
- *Exploratory factor analysis (EFA):* An EFA identified latent factors and applied strict criteria, including eigenvalues >1 and factor loadings $\geq .3$, ensuring construct validity.
- *Correlation and regression analysis:* Relationships among variables were tested using Pearson correlations and multiple regression analysis. Hypotheses were validated through an analysis of variance and t tests, confirming the robustness and statistical significance of findings.

RESULTS

Measurement Model and Metric Invariance

Reliability testing of the scale through a Cronbach's α coefficient aimed to exclude variables with total correlation coefficients $\leq .3$. The criterion for selecting a scale is when the Cronbach's α reliability coefficient is $\geq .6$. A scale with a Cronbach's α reliability coefficient $\geq .6$ was also chosen when used for the first time (Nunnally & Bernstein, 1994). Theoretically, the higher the Cronbach's α , the better. The results of the Cronbach's α test for the component scales are presented in Table 2.

Table 2. Reliability and convergent validity of the measurement scales

Construct	Items	Cronbach's α	Scale mean if item deleted	Scale variance if item deleted	Corrected item-total correlation	Cronbach's α if item deleted
Performance expectancy (PE)	PE1—Online shopping helps save time.	.823	11.97	5.369	.671	.766
	PE2—Online shopping makes it easy to buy products anytime or anywhere.	.823	12.06	5.637	.618	.790
	PE3—Online shopping helps find effective product information easily.	.823	11.97	4.912	.695	.754
	PE4—Online shopping helps save on transportation costs.	.823	11.64	5.484	.606	.795
Social influence (SI)	SI1—Comments and reviews from acquaintances such as friends or relatives influence me when shopping online.	.791	8.51	1.972	.710	.631
	SI2—Comments and reviews from idols or influencers influence me when shopping online.	.791	8.50	2.130	.588	.762
	SI3—Comments and reviews from previous buyers influence me when shopping online.	.791	8.92	2.121	.602	.748
Price value (PV)	PV1—Online prices are cheaper than market prices.	.779	8.03	2.410	.504	.822
	PV2—Shopping online has many price incentives, such as discount vouchers and earning coins or points for products and services.	.779	8.34	2.036	.688	.617
	PV3—Prices are easy to compare when shopping online.	.779	8.27	2.283	.668	.649
Hedonic motivation (HM)	HM1—I derive satisfaction from accumulating points to receive discounts on future purchases when shopping.	.829	7.19	2.940	.706	.750
	HM2—I derive satisfaction from mini-games or livestreams offering tangible rewards (coins or shopping vouchers).	.829	7.96	3.485	.677	.777
	HM3—I derive satisfaction from purchasing unique and novel products.	.829	7.48	3.367	.689	.764
Long-Tail effect (LT)	LT1—I perceive a diverse range of product designs available for purchase.	.818	8.36	2.940	.614	.810
	LT2—I perceive a wide range of brands available when shopping.	.818	8.31	3.173	.665	.762
	LT3—Electronic word-of-mouth (eWOM) increases the frequency of my online shopping.	.818	8.45	2.535	.750	.666
eWOM	eWOM1—eWOM increases the effectiveness of online shopping.	.799	8.13	2.401	.630	.747
	eWOM2—eWOM enhances the trustworthiness of online shopping.	.799	8.10	2.632	.630	.742
	eWOM3—eWOM increases the frequency of online shopping.	.799	8.34	2.641	.680	.693

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Table 2. Continued

Construct	Items	Cronbach's α	Scale mean if item deleted	Scale variance if item deleted	Corrected item-total correlation	Cronbach's α if item deleted
Perceived risk (PR)	PR1—I am concerned that the actual product quality may not meet expectations when shopping.	.839	7.17	7.998	.773	.768
	PR2—I am concerned that the product may not be delivered on time or may be damaged during shipping when shopping.	.839	7.26	8.840	.722	.788
	PR3—I am concerned about the return/refund policy when shopping.	.839	7.12	8.419	.593	.822
	PR4—I am concerned about low personal information security when shopping.	.839	6.90	9.447	.468	.852
	PR5—I am concerned about the safety of the online payment system when shopping.	.839	7.17	8.521	.687	.793
Purchase intentions (PI)	PI1—I will continue to use online shopping service.	.735	7.74	1.891	.640	.560
	PI2—I intend to shop online in the future.	.735	7.72	1.984	.640	.569
	PI3—I plan to shop online in the future.	.735	7.87	1.863	.436	.826

The performance expectancy scale, which assessed users' expectations about products/services, exhibited high reliability, with a Cronbach's α coefficient of .823, exceeding the threshold of .6. Similarly, the social influence scale demonstrated reliability, with a Cronbach's α coefficient of .791. The price value scale, which evaluated perceived value for the money spent, also met requirements, with a Cronbach's α coefficient of .779. Furthermore, the hedonic motivation scale, which captured enjoyment motivation, achieved reliability, with a Cronbach's α coefficient of .829. The long-tail effect scale showed reliability, with a Cronbach's α coefficient of .818.

In addition, the eWOM scale proved reliable, with a Cronbach's α coefficient of .799. The perceived risk scale achieved reliability, with a Cronbach's α coefficient of .839. Finally, the purchase intentions scale met reliability standards, with a Cronbach's α coefficient of .735.

All scales display high reliability based on Cronbach's α and total correlation coefficients, indicating their effectiveness and stability. With eight factors, including seven independent variables and one dependent variable, 27 observed variables were retained, fulfilling the initial hypothesis related to conducting an EFA.

EFA

An EFA was used to examine the construct validity of the measurement model by identifying underlying factors, reducing data complexity, and confirming the convergence and discriminant validity of the observed variables (Trong & Ngoc, 2008). Key indicators used in the analysis include the Kaiser–Meyer–Olkin measure of sampling adequacy and Bartlett's Test of Sphericity, which assess whether the data are suitable for factor analysis.

The Kaiser–Meyer–Olkin (KMO) measure of sampling adequacy was 0.768, exceeding the recommended threshold of 0.6, and Bartlett's Test was significant, $\chi^2(276) = 2,879.343$, $p < .001$, confirming that the dataset was appropriate for factor extraction.

A principal-components analysis with Varimax rotation was conducted on 24 observed variables. As shown in Table 3, seven distinct factors emerged, corresponding to the hypothesized constructs: Performance Expectancy, Social Influence, Price Value, Hedonic Motivation, Long-Tail Effect, eWOM, and PR. All items loaded significantly (factor loading ≥ 0.587), with no cross-loading, indicating strong convergent and discriminant validity.

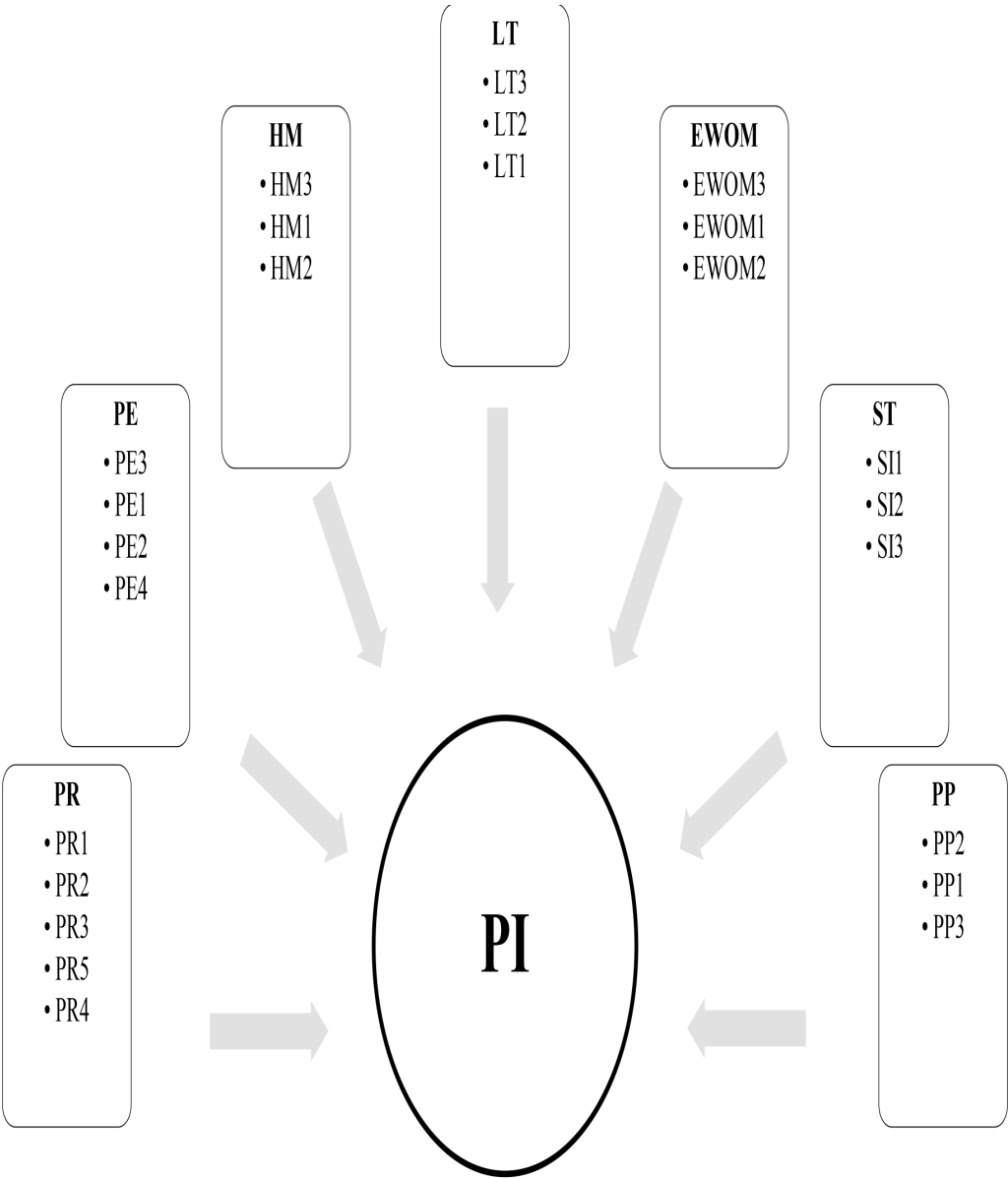
From the EFA analysis results given in Table 3 one can see that all observed variables have been correctly grouped into the initial 7 hypotheses. There is no merging or splitting of variables across different factor groups, and all variables have distinctive values. However, there are changes in the order of arrangement of the factors, which are depicted in Figure 3.

Table 3. Rotated component matrix

Component	1	2	3	4	5	6	7
PR1	.854						
PR2	.853						
PR3	.780						
PR5	.773						
PR4	.587						
PE3		.837					
PE1		.783					
PE2		.777					
PE4		.735					
HM3			.823				
HM1			.822				
HM2			.754				
LT3				.892			
LT2				.847			
LT1				.795			
EWOM3					.868		
EWOM1					.803		
EWOM2					.788		
SI1						.875	
SI2						.798	
SI3						.766	
PP2							.843
PP1							.775
PP3							.769

Note. Extraction method: principal-components analysis. Rotation method: Varimax with Kaiser normalization. The rotation converged in six iterations. PR = Perceived Risk; PE = Performance Expectancy; HM = Hedonic Motivation; LT = Long-Tail Effect; EWOM = Electronic Word of Mouth; SI = Social Influence; PV = Price Value.

Figure 3. Modified scale



Note. PR = Perceived Risk; PE = Performance Expectancy; HM = Hedonic Motivation; LT = Long-Tail Effect; EWOM = Electronic Word of Mouth; SI = Social Influence; PV = Price Value.

After the EFA of the independent variables was conducted, a separate factor analysis was conducted on the dependent variable, Purchase Intentions, which consisted of three items. The Kaiser–Meyer–Olkin (KMO) value was 0.628, exceeding the minimum threshold of 0.60, and Bartlett’s test of sphericity was significant, $\chi^2(3) = 236.17$, $p < .001$, confirming the adequacy of the sample for factor extraction.

The results (see Table 4 and Table 5) indicate a single factor with an eigenvalue >1, explaining 67.4% of the variance, and all items exhibited high loadings (≥ 0.692), demonstrating strong construct validity.

Table 4. Total variance explained

Component	Initial eigenvalues			Extraction sums of squared loadings		
	Total	% of Variance	Cumulative %	Total	% of Variance	Cumulative %
1	2.022	67.403	67.403	2.022	67.403	67.403
2	.683	22.753	90.156			
3	.295	9.844	100.000			

Note. Extraction method: principal-components analysis.

Table 5. Component matrix

Component	
PI1	.879
PI2	.877
PI3	.692

Note. Extraction method: principal-components analysis. One component was extracted. PI = Purchase Intentions.

Multivariate Linear Regression Analysis

After the Cronbach’s α analysis and EFA were conducted, the research model remained unchanged from the original, with seven independent variables—(a) performance expectancy, (b) social influence, (c) perceived value, (d) hedonic motivation, (e) long-tail effect, (f) eWOM, and (g) perceived risk—all of which affect consumers’ intentions to make online purchases.

Correlation and regression analyses were then applied, yielding the following standardized beta coefficients for the regression model given in Equation 1:

$$PI = 0.217 \times PE + 0.210 \times SI + 0.113 \times PV + 0.278 \times HM + 0.150 \times LT + 0.318 \times eWOM - 0.144 \times PR, \quad (1)$$

in which PI = purchase intentions, PE = performance expectancy, SI = social influence, PV = perceived value, HM = hedonic motivation, LT = long-tail effect, and PR = perceived risk.

The independent variables are expectancy, social influence, perceived value, hedonic motivation, long-tail effect, eWOM, and perceived risk. The dependent variable was the purchase intentions of online consumers.

Table 6. Regression coefficient results

Model 1	Unstandardized coefficients		Standardized coefficients	<i>t</i>	<i>p</i>	Correlations			Collinearity statistics	
	B	SE				Zero order	Partial	Part	Tolerance	VIF
(Constant)	-.310	.251		-1.233	.219					
PE	.188	.034	.217	5.593	.000	.489	.329	.199	.835	1.197
SI	.199	.036	.210	5.506	.000	.455	.324	.195	.867	1.154
PV	.103	.036	.113	2.845	.005	.414	.174	.101	.800	1.249
HM	.208	.031	.278	6.616	.000	.570	.381	.235	.715	1.398
LT	.120	.030	.150	4.044	.000	.312	.244	.144	.920	1.087
EWOM	.271	.033	.318	8.168	.000	.566	.453	.290	.832	1.202
PR	-.129	.034	-.144	-3.838	.000	-.288	-.232	-.136	.901	1.109

Note. Dependent variable: purchase intentions (PI). VIF = variance inflation factor; PE = performance expectancy; SI = social influence; PV = price value; HM = hedonic motivation; LT = long-tail effect; EWOM = electronic word of mouth; PR = perceived risk.

This study explored the e-commerce landscape in high-cost urban areas, focusing on the online shopping behavior of consumers in Hanoi. The key findings revealed that performance expectancy ($\beta = 0.217$, $p < .05$) and social influence ($\beta = 0.210$, $p < .05$) significantly drive online purchase intentions. Perceived value ($\beta = 0.113$, $p < .05$), hedonic motivation ($\beta = 0.278$), the long-tail effect ($\beta = 0.150$), and eWOM ($\beta = 0.318$) also positively affected consumers' intentions. Conversely, perceived risk ($\beta = -0.144$) negatively influences online shopping intentions, highlighting it as a notable obstacle (Table 6). Therefore, all hypotheses are accepted (Table 7).

Table 7. Hypothesis testing results

Hypothesis (H)	Description	<i>p</i>	VIF	Result
H1	Efficiency expectation positively influences consumers' intentions to shop online.	.000	1.197	Accepted
H2	Social influence positively influences consumers' intentions to shop online.	.000	1.154	Accepted
H3	Price value positively influences consumers' intentions to shop online.	.005	1.249	Accepted
H4	Hedonic motivation positively influences consumers' intentions to shop online.	.000	1.398	Accepted
H5	The long-tail effect positively influences consumers' online shopping intentions.	.000	1.087	Accepted
H6	Electronic word of mouth positively influences consumers' online shopping intentions.	.000	1.202	Accepted
H7	Perceived risk negatively influences consumers' intentions to shop online.	.000	1.109	Accepted

HOW INCOME SHAPES ONLINE SHOPPING INTENTIONS

To further explore how socioeconomic status shapes online shopping behavior in a high-cost urban context, the study tested Hypothesis 8, which posited that there is a significant difference in online shopping intentions among consumers across different income groups.

A one-way analysis of variance was conducted to assess whether mean purchase intention scores differed significantly between income categories. Before the test, Levene's Test of Homogeneity of Variance was performed to verify the assumption of equal variances (see Table 8). The result ($F =$

3.639, $p = .013$) indicates a violation of the homogeneity assumption; therefore, the Welch test, which is robust to unequal variances, was used (see Table 9).

Table 8. Test of homogeneity of variances by income group

PI	Levene statistic	df1	df2	p
Based on mean	3.639	3	262	.013
Based on median	1.825	3	262	.143
Based on the median and with adjusted df	1.825	3	252.584	.143
Based on trimmed mean	3.268	3	262	.022

Note. PI = purchase intentions.

Table 9. Robust tests of equality of means: Purchase intentions (PI)

Test	Statistic ^a	df1	df2	p
Welch	6.558	3	55.369	.001
Brown–Forsythe	5.646	3	104.049	.001

^aAsymptotically F distributed.

The Welch test result confirms that significant differences exist in purchase intentions across income groups. To identify where the differences lie, Tamhane’s Time 2 post hoc test was conducted because of unequal variances. The results (see Table 10) show that the <USD 80 group reported significantly higher online shopping intentions compared with the USD 80–200 group (mean difference = .39485, $p = .001$). No statistically significant differences were observed between other pairs of income groups.

Table 10. Post hoc tests: Analysis of variance by income

Thunhap (I)	Thunhap (J)	Mean difference (I – J)	SE	p	95% CI
<USD 80	USD 80–200	.39485*	.09287	.001	[.1389, .6508]
	USD 200–400	.18841	.14939	.769	[-.2318, .6086]
	>USD 400	.13979	.12141	.827	[-.1894, .4690]
USD 80–200	<USD 80	-.39485*	.09287	.001	[-.6508, -.1389]
	USD 200–400	-.20644	.13710	.610	[-.5993, .1864]
	>USD 400	-.25505	.10592	.106	[-.5411, .0310]
USD 200–400	<USD 80	-.18841	.14939	.769	[-.6086, .2318]
	USD 80–200	.20644	.13710	.610	[-.1864, .5993]
	>USD 400	-.04861	.15783	1.000	[-.4865, .3893]

continued on following page

Table 10. Continued

Thunhap (I)	Thunhap (J)	Mean difference (I – J)	SE	p	95% CI
>USD 400	<USD 80	-.13979	.12141	.827	[-.4690, .1894]
	USD 80–200	.25505	.10592	.106	[-.0310, .5411]
	USD 200–400	.04861	.15783	1.000	[-.3893, .4865]

Note. Multiple comparisons conducted using Tamhane's T2 test. Dependent variable: purchase intentions. CI = confidence interval.

*Mean difference is significant at $p < .05$.

Descriptive statistics further revealed that the lowest income group (<USD 80) reported the highest mean score ($M = 4.19$) for online shopping intentions, followed by the USD 200–400 and >USD 400 groups, whereas the USD 80–200 group reported the lowest mean score ($M = 3.79$).

Table 11. Purchase intention differences by income

Income	N	M	SD	SE	95% CI	Min	Max	Between- component variance
<USD 80	23	4.1884	.37371	.07792	[4.0268, 4.3500]	3.33	5.00	
USD 80–200	176	3.7936	.67023	.05052	[3.6939, 3.8933]	2.00	5.00	
USD 200–400	19	4.0000	.55556	.12745	[3.7322, 4.2678]	2.67	5.00	
>USD 400	48	4.0486	.64500	.09310	[3.8613, 4.2359]	2.67	5.00	
Total	266	3.8885	.64986	.03985	[3.8100, 3.9669]	2.00	5.00	
Model								
Fixed effects			.63844	.03915	[3.8114, 3.9656]			
Random effects				.12338	[3.4958, 4.2811]			.02835

Note. CI = confidence interval; Min = minimum; Max = maximum.

These findings reveal a noteworthy deviation from common expectations: Consumers in the lowest income group (<USD 80) reported significantly higher online shopping intentions compared with those in the USD 80–200 range. Although not all pairwise comparisons reached statistical significance, the pattern suggests that lower income consumers may demonstrate greater motivation to use, or reliance on, online shopping channels than middle-income groups.

DISCUSSION

The adoption of an enhanced UTAUT2 model confirmed that factors such as eWOM, hedonic motivation, performance expectancy, social influence, the long-tail effect, and price value are positively correlated with online shopping intentions, which is consistent with prior research (Cheng et al., 2002; Davis, 1985; Singh & Matsui, 2017; Venkatesh et al., 2012; Zhang et al., 2023). Perceived risk is the only factor that has a negative influence (Almousa, 2014; Lin et al., 2020; Sudibyo et al., 2020). These results reinforce the theoretical model by empirically validating the directional influence of all seven hypothesized factors.

Among these determinants, eWOM emerged as the most influential driver, with a standardized beta coefficient of 0.318 ($p < .001$). This finding is consistent with results found by Cheung and Thadani (2012) and Lee and Youn (2009), who emphasized its critical role in fostering consumer trust. In Hanoi's high-cost urban market, where consumers are highly sensitive to financial risks, trustworthy reviews and peer recommendations are particularly significant, amplifying the role of eWOM. Nguyen et al. (2019) and Bender and Mattoso (2022) similarly have identified eWOM as a vital trust-building tool in Vietnam's urban settings. The present study expands on the existing literature by demonstrating how eWOM enhances not only perceived reliability but also the frequency and effectiveness of online shopping. These findings also are consistent with recent regional reports highlighting how social commerce and peer influence increasingly shape consumer behavior across Southeast Asia (Yendamuri et al., 2023).

The long-tail effect, with a standardized beta coefficient of 0.150 ($p < .001$), illustrates the important role of niche product offerings in influencing online shopping behavior. Anderson (2006) and Brynjolfsson et al. (2011) have argued that e-commerce platforms benefit from curating diverse product selections that accommodate unique consumer preferences. In Hanoi, increasing digital literacy and platform accessibility are driving the demand for product variety, in particular among younger and cost-conscious users. Prior studies (Arunachalam et al., 2020; Pels & Sheth, 2017) also have emphasized the importance of variety and facilitating conditions in engaging low-income consumers in emerging markets. These findings resonate with broader trends across the Asia-Pacific region, where consumers increasingly turn to digital platforms to discover value in niche and cross-border product categories (Arora et al., 2025).

The negative effect of perceived risk is consistent with classical consumer behavior theories, which propose that heightened perceptions of financial, product, and privacy risks diminish consumers' willingness to make purchases (Ariffin et al., 2018; Ugochukwu et al., 2021). In Vietnam's urban markets, risk perception remains a prominent barrier, especially among older consumers and those with limited digital literacy (Phuong & Chotchakornpant, 2024). This confirms that reducing uncertainty and enhancing platform security are crucial steps for increasing e-commerce participation. Similar findings have emerged in Malaysia and Indonesia, where trust and privacy concerns have significantly constrained e-commerce growth despite rising platform usage (PWC, 2023).

An unexpected finding relates to income-based differences in purchase intentions. Consumers earning <USD 80 per month demonstrated higher online shopping intentions compared with higher income groups. This challenges traditional assumptions linking higher disposable income with greater online activity (Dong et al., 2020; Zhao & Wang, 2015). Despite well-documented participation barriers and negative attitudes among low-income users (Bender & Mattoso, 2022), these groups may perceive e-commerce as a viable channel to access affordable goods. Although concerns around security, delivery, and affordability persist, the perceived trust enhancement provided by eWOM appears to offset some of these limitations (Dakduk et al., 2020). Moreover, financial incentives, such as free shipping and product-level discounts, can significantly boost intention to buy (Rummo et al., 2023). This pattern echoes emerging insights from Indonesia and Thailand, where low-income consumers increasingly engage in online shopping when platforms offer promotions and simplified user experiences (Yendamuri et al., 2023).

Taken together, these insights point to a more complex set of motivations that are shaping online shopping intentions in high-cost urban environments, rather than being driven solely by access or income level. To be specific, the findings clearly indicate that e-commerce effectively serves as a viable alternative to traditional shopping, enabling lower income consumers in particular to overcome economic barriers and access affordable goods and services despite rising living costs. Recent observations from Hanoi's traditional markets, which report declining foot traffic and difficulty competing with online alternatives, further underline this shift (Hien, 2023; Vũ, 2024). Thus, by leveraging trust-building mechanisms, such as eWOM and targeted economic incentives, e-commerce platforms can address unique challenges faced by consumers in high-cost urban areas, promoting

broader participation and inclusion. The sustained growth of e-commerce in these markets ultimately may rely less on infrastructural developments and more on how effectively platforms interpret and respond to evolving consumer sensitivities.

CONCLUSION

Set against the backdrop of Hanoi's dense, high-cost urban environment, this study offers a focused perspective on how online shopping intentions are shaped under economic and behavioral constraints. Rather than reinforcing broad generalizations about digital consumption, the findings reveal specific mechanisms—trust cultivated through eWOM, utility perceived through efficiency, and relevance supported by product diversity—that consistently influence consumer behavior.

By incorporating eWOM and the long-tail effect into the UTAUT2 framework, this study refines our understanding of online shopping intention as a context-sensitive outcome shaped by affordability, perceived control, and social validation. The consistently high intention noted among lower income consumers is particularly notable. It challenges assumptions about digital exclusion and highlights how perceived value, more than income level alone, motivates participation in e-commerce. It is important to note that these insights support the proposition that e-commerce can serve as a viable alternative to traditional shopping formats in high-cost urban environments, in particular for economically constrained consumers.

From a theoretical standpoint, this study contributes to extending technology acceptance research into socioeconomically complex settings. The adapted model emphasizes the importance of integrating local context into established frameworks, especially in emerging Southeast Asian markets, where affordability and platform trust play critical roles. In doing so, this study adds empirically grounded insights to ongoing discussions about how e-commerce operates at the intersection of consumer behavior, socioeconomic constraints, and digital opportunity.

On a practical level, these findings emphasize the need for platform strategies that go beyond universal design. Features such as verified reviews, accessible interfaces, and meaningful promotions not only enhance user experience but also serve as entry points for broader consumer engagement. In high-cost cities, strategies aligned with financial sensitivity and trust building are more likely to reach diverse consumer groups.

From a policy and business perspective, these findings have important implications for both e-commerce platforms and government stakeholders. The significant role of trust-related variables, such as eWOM, suggests that platforms should prioritize investments in credibility-enhancing mechanisms, such as verified user reviews, transparent seller ratings, and responsive customer service systems. These mechanisms are especially crucial in high-cost urban environments, where financial caution amplifies sensitivity to risk.

In addition, the long-tail effect underscores the strategic value of offering diverse, niche product categories that cater to various consumer segments. For platforms operating in dense urban markets, personalization becomes more than a competitive feature—it serves as a practical means of engaging consumers with varying preferences and budget constraints.

From a public policy standpoint, the consistently strong shopping intentions among lower income consumers highlight the importance of inclusive digital strategies. Policies that improve access to infrastructure, promote digital literacy, and facilitate mobile payment adoption can ensure that e-commerce serves not only as a commercial channel but also as a tool for broader digital inclusion. Public-private collaborations could further foster safer, more accessible, and context-sensitive e-commerce platforms.

Although these findings are grounded in one urban setting, their broader relevance suggests opportunities for comparative research. Future studies could explore how digital inclusion, sustainability preferences, or platform transparency influence online shopping behavior in other high-density urban markets.

FUNDING STATEMENT

This research is funded by Thuongmai University, Hanoi, Vietnam.

COMPETING INTERESTS

The author of this publication declares there are no competing interests regarding the publication of this manuscript.

PROCESS DATES

04, 2025

This manuscript was initially received for consideration for the journal on 01/16/2025, revisions were received for the manuscript following the double-anonymized peer review on 03/24/2025, the manuscript was formally accepted on 03/28/2025, and the manuscript was finalized for publication on 04/28/2025

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